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RESEARCH

A novel hybrid graph neural network and transformer model for intrusion detection

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Abstract

The increasing complexity and diversity of cyber-attacks in interconnected systems, including the Internet of Things (IoT), industrial control networks, and critical infrastructure, have exposed the limitations of traditional intrusion detection methods based on predefined rules or static architectures. To address these challenges, we propose a novel hybrid intrusion detection framework that synergistically integrates Graph Neural Networks (GNNs) and Transformer architectures through a dual-domain feature fusion strategy. Specifically, network traffic data are converted into grayscale images, from which texture features are extracted via the Gray Level Co-occurrence Matrix (GLCM) method and fused with traffic-based features to enhance spatial–temporal representation. Borderline-Synthetic Minority Over-sampling Technique (SMOTE) is employed to mitigate class imbalance, improving robustness against minority-class misclassification. Furthermore, a Kolmogorov–Arnold Network (KAN) replaces the conventional multilayer perceptron (MLP) head to capture richer nonlinear decision boundaries. Extensive experiments on UNSW_NB15, IEC61850, and DAPT2020 datasets show that our approach consistently outperforms state-of-the-art baselines in accuracy, achieving up to 99.24% while maintaining competitive efficiency. The proposed framework offers a scalable and generalizable solution for intrusion detection across diverse network environments, contributing to the advancement of resilient cybersecurity systems.

Keywords Intrusion detection · Cyber-physical system · Gray level co-occurrence matrix · Graph neural network

1 Introduction

Cybersecurity threats have become increasingly sophisticated and prevalent with the widespread adoption of interconnected systems, including Internet of Things (IoT), industrial control systems, and general network infrastructures. These interconnected networks facilitate seamless integration of computation, communication, and control technologies [1]. However, their intrinsic openness and interconnectivity also significantly amplify their exposure to malicious intrusions, posing severe threats to network reliability, data integrity, and overall system security. Fig. 1 shows the framework of power cyber-physical systems.

Real-world incidents underline the importance of robust cybersecurity frameworks. Examples such as the 2016 Mirai botnet attack, which disrupted global IoT infrastructures, and multiple high-profile data breaches highlight the critical need for efficient intrusion detection systems (IDS) capable of early threat detection and mitigation [2]. Traditional cybersecurity measures, predominantly relying on firewalls and perimeter security, have proven inadequate against evolving cyber threats. With the increasing number of distributed devices having limited computing resources, deploying complex encryption algorithms becomes impractical, leaving networks vulnerable to sophisticated attacks such as denial-of-service (DoS), replay attacks, and data tampering. These threats necessitate advanced detection mechanisms that go beyond static rule-based systems.

Data-driven intrusion detection methods have emerged as promising solutions, addressing some limitations inherent in conventional rule-based systems. Especially in complex real systems, Fig. 1 shows the vision of intrusion detection using the power cyber-physical system as an example. Classical machine learning approaches, including

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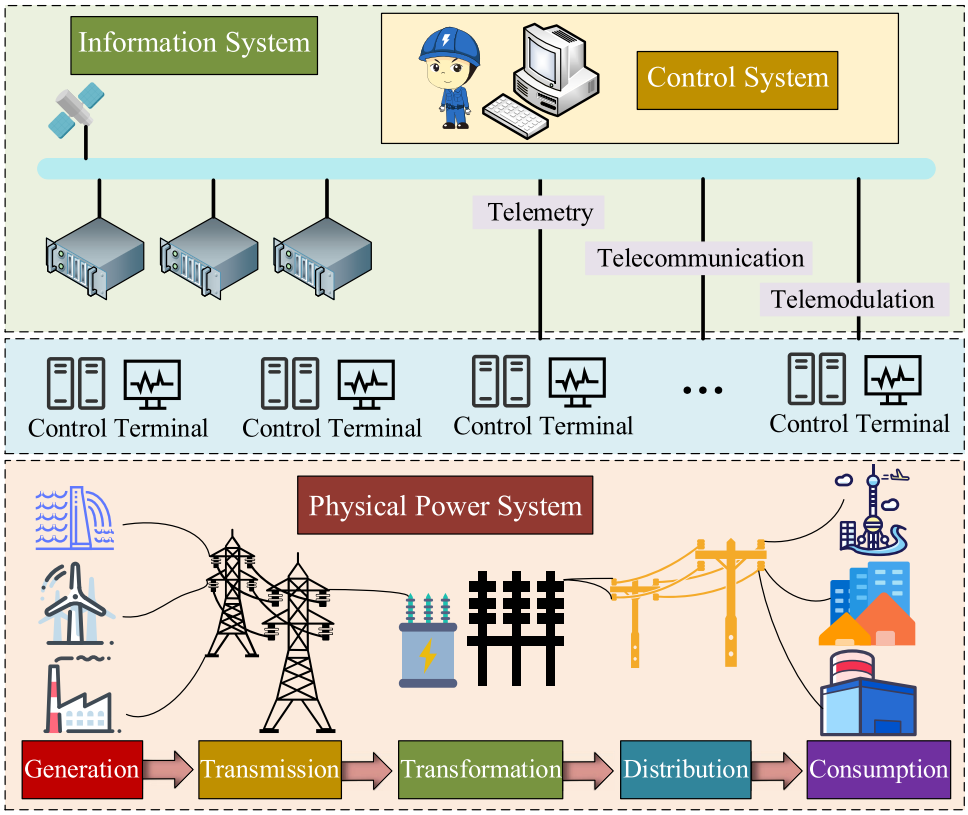


Fig. 1 The cyber-physical system is divided into two main components: the Physical Power System and the Information System. These two systems are interconnected through Control Terminals such as Telemetry, Telecommunication, and Telemodulation. The abnormal traffic detection and reporting process is illustrated as follows: (1) Network traffic data is collected from the physical power system via control terminals; (2) The data is analyzed using intrusion detection system, which processes the traffic and identifies any anomalies; (3) When abnormal traffic is detected, an anomaly report is generated and sent to the central controller via a secure communication channel; (4) The central controller discards attack packets and updates the flow table to maintain system security and integrity

Decision Trees (DT), Random Forests (RF), and Logistic Regression (LR), have been extensively applied [3]. However, these models often struggle with high-dimensional, heterogeneous data and fail to effectively model intricate feature relationships, particularly in multi-class classification scenarios. Furthermore, traditional methods typically lack robust mechanisms for handling data imbalance, which diminishes their detection performance [4]. Recent advancements in deep learning, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have significantly improved intrusion detection accuracy by efficiently capturing sequential and spatial patterns in data [5, 6]. Nevertheless, these models still encounter difficulties in capturing diverse feature representations and preventing the over-smoothing problem prevalent in deeper network architectures [7]. Additionally, conventional deep learning models are less effective at modeling non-Euclidean, graph-structured relationships inherent in network traffic data.

Most existing IDS frameworks extract either inter-packet topology or raw-byte texture but rarely both. This single-view design leaves blind spots when multi-stage, graph-structured attacks evolve across protocol layers. Therefore, we explicitly ask: Can a unified detector fuse topology and texture so that both long-range dependencies and low-level payload cues are retained? Graph Neural Networks (GNNs) excel at modeling non-Euclidean structures by representing data as graphs, effectively capturing complex interdependencies between network traffic features [8]. Transformers, characterized by their powerful multi-head attention mechanisms, have demonstrated remarkable capabilities in handling long-range dependencies and complex feature interactions, crucial for accurate multi-class classification [9]. Despite their strengths, combining GNNs and Transformers efficiently remains an open challenge, particularly in effectively integrating diverse feature representations and enhancing the overall robustness of intrusion detection models. Recent hybrids such as ResDNViT [10] and ET-GraphSAGE [11] represent first attempts at combining vision and graph models, yet they treat packet images and graph links as loosely coupled features.

To address these limitations, we propose a novel hybrid intrusion detection framework that synergistically integrates GNN and Transformer architectures. Specifically, our model introduces an innovative fusion mechanism by converting network traffic data into grayscale images, from which texture features are extracted using Gray Level Co-occurrence Matrix (GLCM). These texture features are strategically fused with conventional traffic features, significantly

enriching the feature representation. Furthermore, we construct graphs based on image patches derived from traffic data images, allowing GNNs to effectively model spatial relationships. To address the inherent limitations in conventional architectures, we employ the Kolmogorov-Arnold Network (KAN) as a unique output layer, enhancing the model's capability in capturing complex, nonlinear relationships between features.

The primary contributions of this paper are summarized as follows:

- We design the novel hybrid IDS that jointly exploits GLCM byte-texture and patch-level graph topology within a single computation graph, closing the long-standing gap between non-Euclidean inter-packet relations and low-level payload cues.
- By introducing a lightweight KAN output layer atop the GNN–Transformer blocks, the model alleviates Transformer over-smoothing and captures highly nonlinear cross-view dependencies without extra depth.
- Extensive experimental evaluations on diverse benchmark datasets, including UNSW_NB15, IEC61850, and DAPT2020, demonstrate that our approach significantly outperforms state-of-the-art methods, achieving an accuracy of 99.24%.

The remainder of this paper is organized as follows: Section 2 reviews related research in data-driven intrusion detection methods. Section 3 details the proposed GNN-Transformer model, including data preprocessing and feature extraction methodologies. Section 4 presents the experimental setup and comprehensive performance evaluations. Finally, Section 5 concludes the paper and discusses potential directions for future research.

2 Related work

Machine learning algorithms have increasingly become pivotal in enhancing cybersecurity, particularly within IDS. Traditional machine learning approaches such as Support Vector Machines (SVM) [12] and DT [13], while initially effective, require careful tuning and optimization to achieve high detection accuracy and generalization to novel attack patterns. Ensemble learning methods, including RF and deep random forests, have been proposed to overcome these limitations. Mogal and Joshi [14] combined RF with Penalized Attribute Forests, identifying attacks through majority voting mechanisms, while Hu et al. [15] developed a dynamic deep random forest that enhanced detection through cascaded feature extraction and classification processes.

Addressing high-dimensional data challenges, feature selection techniques have been extensively studied to improve detection performance. For instance, Khan et al. [16] employed a weighted particle swarm optimization method for effective feature extraction, significantly enhancing model performance with RF and neural networks. Similarly, Hossain et al. [17] integrated multiple algorithms for feature selection, and Jeffrey et al. [18] applied CNNs, showing superior detection accuracy across various datasets.

Imbalanced datasets pose significant challenges for machine learning models. Techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) [19] have been introduced to generate synthetic minority class samples, thereby balancing data distributions. Advanced approaches combining oversampling and undersampling [20] and safe sample selection using k-means clustering [4] further improved model robustness by minimizing the impact of outlier synthetic samples.

Despite advancements, traditional deep learning models like CNNs and Long Short-Term Memory networks (LSTMs) are primarily suited for Euclidean data structures, limiting their capability in capturing non-Euclidean, graph-based relationships inherent in network traffic data. GNNs have emerged to address this limitation by effectively modeling complex topological patterns in network traffic [21]. However, deep GNN architectures face challenges such as over-smoothing, which leads to reduced feature diversity [7].

Recent studies have begun combining GNNs with Transformer architectures, leveraging the complementary strengths of GNNs in modeling spatial dependencies and Transformers in capturing long-range temporal relationships through multi-head self-attention mechanisms [22]. This hybrid approach has shown promising results in various classification tasks but remains underexplored in general intrusion detection scenarios. In addition to these advancements, recent research has proposed novel methods for enhancing intrusion detection. One such method is a hybrid Graph Neural Network and Transformer Encoder (GTE) approach for anomaly and cyber threat detection in smart grids. This method utilizes GNN for spatial feature extraction based on the power system topology and Transformer encoder to capture long-range dependencies. The approach, which was evaluated on the PMU dataset, achieved high accuracy (97.90%) and outperformed models like Gaussian Mixture Model (GMM) and Spatial Pyramid Pooling with CNN [23]. Similarly, ET-GraphSAGE (ETG) integrates GNN and Transformer techniques to address the complexities of network intrusion detection. The model captures relationships between network nodes with GNN and enhances its handling of

long-range dependencies in streaming network data using Transformer. This method demonstrated broad applicability and robustness, outperforming existing techniques across several datasets, including BoT-IoT and NF-BoT-IoT [11]. *Beyond GNN–Transformer hybrids*, an Edge-featured Multi-hop Attention GNN (EMA-IDS) further explores graph topology by propagating multi-hop attention and simultaneously encoding node and edge attributes, thereby mitigating over-smoothing and improving computational efficiency. EMA-IDS yielded superior results on four public benchmarks, including NF-CSE-CIC-IDS2018-v2 and NF-UNSW-NB15-v2 [24]. Building on the momentum of attention mechanisms, the attention-based Transformer (AT-IDS) has been proposed for IoT networks, where self-attention is leveraged to model complex traffic dependencies and evaluated on NSL-KDD and UNSW-NB15, the model exceeded 99% in accuracy, precision and recall for both binary and multiclass settings [25]. Extending the vision-transformer paradigm to NetFlow analysis, *ResDNViT* combines a self-attention ViT encoder with a lightweight residual-dense network. By reshaping NetFlow features into 2-D patches, the model achieved state-of-the-art accuracy and strong cross-device generalisation on CICIDS2017_improved, Bot-IoT, CICIoT2022 and N-BaIoT datasets [10]. In addition, several recent studies have explored secure communication traffic-aware paradigms. SRU-Net with Dynamic Behavior Aggregation [26] reduces client traffic through adaptive participation, Federated-Boosting [27] addresses data imbalance in the consumer IoT through a weighted boosting algorithm, and Fed-Inforce-Fusion [28] combines Q-learning with federated fusion for the medical IoT. While these studies improve scalability and privacy, they still process topology and payload cues in separate feature spaces, leaving cross-view synergy unexplored.

Distinctiveness of the present work Summarising the above survey, prior efforts fall into five methodological streams. *First*, classical or ensemble learners (e.g., SVM and RF [12, 14]) depend on hand-crafted statistics and therefore overlook rich payload semantics. *Second*, feature-selection and imbalance remedies such as SMOTE variants [4, 19] alleviate minority mis-classification yet leave the single-view representation unchanged. *Third*, topology-centric GNN IDSs (GCN-IDS [21], TGAT [7]) model non-Euclidean links but suffer from over-smoothing and ignore byte-level texture. *Fourth*, texture/sequence-centric deep networks (CNN, RNN and Transformer only [5, 6]) excel at local semantics but treat packets as isolated images or sequences, omitting cross-flow relations. *Fifth*, the latest hybrids ResDNViT [10], ET-GraphSAGE [11], EMA-IDS [24] and AT-IDS [25]—serially couple vision and graph, so real interaction occurs only in a shallow fusion layer and GNN over-smoothing remains unresolved. In sharp contrast, our model integrates GLCM textures and patch-level graphs within a *single* embedding space, augments cross-view synergy through a lightweight Kolmogorov–Arnold head that suppresses over-smoothing, and weaves Borderline-SMOTE into training to maintain balanced learning signals. These design choices jointly bridge the representation, modelling and imbalance gaps identified in all five streams.

3 The proposed scheme

3.1 Design principles and justifications

This subsection elaborates on the rationale behind key methodological choices in our proposed scheme, specifically the adoption of GLCM, the transformation of data into grayscale images, and the utilization of the KAN.[1]

GLCM over CNN-based features The choice of GLCM for texture feature extraction stems from its superior capability in explicitly quantifying spatial relationships and texture patterns within images, which are crucial for distinguishing subtle differences in network traffic data. Compared to CNN-based feature extraction methods, GLCM provides interpretable and consistent texture descriptors (e.g., energy, contrast, homogeneity) that directly correspond to specific structural characteristics of the traffic images, facilitating enhanced anomaly detection through richer spatial representation.

Rationale for grayscale image representation Converting network traffic data into grayscale images significantly enhances the feature representation by leveraging spatial context visually. The spatial structure inherent in grayscale images allows Transformers, with their powerful self-attention mechanisms, to effectively capture long-range dependencies and interactions between distant features. This image-based approach enables the model to uncover hidden patterns that might be overlooked by purely numeric data representations, thereby improving overall detection performance.

Advantage of KAN over traditional layers The integration of the KAN as the final layer addresses the complexity and nonlinearity inherent in network intrusion data. KAN,

[1] Code: https://drive.google.com/file/d/1PkCF0s5Voplb-6Ma4JVBTXZttwT8cup5/view?usp=drive_link

leveraging spline functions in combination with multilayer perceptrons, excels in modeling highly complex, nonlinear relationships, surpassing traditional linear or basic nonlinear layers (such as standard multilayer perceptrons (MLPs)). This capability allows the model to robustly represent intricate feature interactions, significantly enhancing detection accuracy and model robustness against varied and sophisticated attack patterns.

End-to-end feature flow Our framework processes features through four successive stages:

1. *Dual-domain construction*: each 42-dimensional traffic vector is padded and reshaped to an 8×8 grayscale image. GLCM extracts a 20-dimensional texture vector, which is concatenated with the original traffic statistics, forming a fused 62-dimensional feature vector $\mathrm{x}_{\mathrm{fuse}}$.
2. *Graph formation*: the grayscale image is divided into N non-overlapping $P \times P$ patches. Patch embeddings ($\in \mathbb{R}^{P^2}$) serve as node attributes, and k-nearest-neighbour edges are added, producing a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.
3. *GNN block*: multiple graph-convolution layers propagate local information, yielding node features $\mathrm{H}_{\mathrm{GNN}} \in \mathbb{R}^{N \times d}$. A read-out operation (mean pooling) concatenated with $\mathrm{x}_{\mathrm{fuse}}$ produces a holistic vector h_0.
4. *Transformer encoder*: h_0 is treated as a sequence token and processed by L self-attention layers to capture global dependencies, producing h_L. Finally, h_L is fed into the KAN head, which outputs the class probabilities.

This pipeline ensures that low-level spatial textures, statistical traffic cues, local graph structures, and long-range interactions are successively integrated before the final nonlinear decision.

To address the aforementioned issues, we convert network traffic samples into grayscale images, enabling the visualization of network attacks and facilitating their quick identification through image analysis. We then apply the GLCM method to extract texture features from these images. By integrating these texture features with the original traffic features, we can mine additional potential features while retaining the essential characteristics of the initial data, thereby enhancing intrusion detection performance. To tackle the data imbalance phenomenon in intrusion detection datasets, we employ the Borderline-SMOTE method to increase minority class samples, achieving balanced data for multiclass classification tasks. Additionally, we employ a GNN-Transformer model, combining GNNs and Transformer architectures, to improve convergence speed and resolve issues related to local optima. The overall scheme consists of three key components: data preprocessing, feature fusion, and the GNN-Transformer model for classification (Fig. 2).

Require: Normalised traffic sample $\mathbf{x} \in \mathbb{R}^{42}$, patch size P, #GNN layers G, #Trans. layers L
Ensure: Class prediction $\hat{y}$
Stage 1: dual-domain feature construction
Pad $\mathbf{x}$ to 64 dim, reshape to 8×8 grayscale image $\mathbf{I}$
Compute GLCM features $\mathbf{t} \in \mathbb{R}^{20}$; concatenate to form $\mathbf{x}_{\mathrm{fuse}} \in \mathbb{R}^{62}$
Stage 2: graph generation
Split $\mathbf{I}$ into $N = \frac{64}{P^2}$ patches and embed each as node feature
Build k-NN edges to obtain graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$
Stage 3: GNN → Transformer → KAN
for $g = 1$ **to** G **do**
  $\mathbf{H} \leftarrow \textsc{GraphConv}(\mathbf{H}, \mathcal{G})$
end for
$\mathbf{h}_0 \leftarrow \textsc{Pool}(\mathbf{H}) \,\|\, \mathbf{x}_{\mathrm{fuse}}$
for $\ell = 1$ **to** L **do**
  $\mathbf{h}_\ell \leftarrow \textsc{TransformerLayer}(\mathbf{h}_{\ell-1})$
end for
$\hat{y} \leftarrow \textsc{KAN_Head}(\mathbf{h}_L)$
return $\hat{y}$

Fig. 2 Steps of the proposed scheme.

Complexity analysis Let n be the number of traffic samples, N the patches (nodes) per sample, E the edges, d the hidden width, h the attention heads, and L (G) the number of Transformer (GNN) layers.

- *Data pre-processing*: $\mathcal{O}(n \cdot 64)$ for padding and $\mathcal{O}(n \cdot 64 \log 64)$ to compute four GLCMs, negligible versus model inference.
- *Graph formation* : $\mathcal{O}(nN \log N)$ with KD-tree k-NN search; storage $\mathcal{O}(n(N+E))$.
- *GNN block*: $\mathcal{O}(G(Ed + Nd^2))$ when using message-passing with linear projections.
- *Transformer encoder*: $\mathcal{O}(LhN^2d)$ for multi-head attention and $\mathcal{O}(LNd^2)$ for feed-forward sub-layers.
- *KAN head*: $\mathcal{O}(d+s)$, where s is the number of spline knots; $s \ll d$ in our implementation.

For the UNSW_NB15 setting (N=16, $E \approx kN$, k=8, d=128, G=3, L=4, h=8), the end-to-end time complexity is dominated by the attention term $\mathcal{O}(LhN^2d) = \mathcal{O}(4 \times 8 \times 256 \times 128) = \mathcal{O}(10^6)$, which is tractable on a single GPU. Memory usage scales linearly with n and d and quadratically with N inside the Transformer, remaining below 2 GB for batch size 64 on all datasets.

3.2 Data preprocessing

To effectively manage the high dimensionality and class imbalance inherent in network traffic data, our preprocessing pipeline includes feature encoding, normalization, data augmentation, and grayscale image conversion for texture feature extraction.

Initially, categorical features such as protocol types ('proto', 'service', 'state') are converted into numerical

representations using Label Encoding. Attributes irrelevant to model performance, such as 'id', and labels ('label', 'attack_cat') are removed from the feature set to prevent data leakage, resulting in a final feature set comprising 42 dimensions.

To standardize the scale of features, we employ logarithmic scaling on features with significantly varied ranges and subsequently apply Min-Max normalization, mapping values within the range [0, 1] using:

$$X = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

To facilitate texture feature extraction, the normalized numerical features are scaled to a grayscale range [0, 255], enabling their representation as pixel intensities in images.

Given the substantial class imbalance within the original UNSW_NB15 dataset, notably underrepresented categories ('Backdoor', 'Shellcode', 'Worms', and 'Analysis'), we apply the Borderline-SMOTE technique to synthetically augment samples near the class boundaries. This method generates additional minority class samples, balancing each category to match the size of the majority class ('Normal'), resulting in a balanced dataset of 560,000 samples. A similar balancing approach is applied to the test set, ensuring even class distribution, as illustrated in Fig. 2.

For binary classification scenarios focusing on anomaly detection, data augmentation is intentionally omitted to prevent potential over-representation of normal samples, preserving model sensitivity to anomalous patterns.

In summary, our comprehensive preprocessing strategy—comprising feature encoding, normalization, Borderline-SMOTE-based augmentation, and grayscale image conversion—ensures robust data preparation, optimizing the training and performance of our hybrid GNN-Transformer intrusion detection model.

3.3 Texture feature fusion with traffic features

To enhance anomaly-detection capability, we implement a feature-fusion strategy that combines texture cues from grayscale images with traditional traffic statistics. Each traffic sample, represented by a 42-dimensional vector after preprocessing, is zero-padded to 64 elements and reshaped into an 8×8 grayscale image. This choice offers a practical advantage: a power-of-two side length produces exactly sixteen non-overlapping 2×2 patches, which simplifies subsequent *k*-nearest-neighbour graph construction and ensures a symmetric neighbourhood for every node. Once the image is formed, GLCM analysis extracts fine-grained texture patterns that are subsequently fused with the numeric traffic features.

The rationale behind converting network traffic data into grayscale images lies in effectively capturing intrinsic spatial patterns that are less discernible in numeric formats. As depicted in Fig. 3, grayscale visualizations of different attack classes exhibit distinct visual patterns, facilitating better differentiation of traffic types.

Utilizing GLCM, we extract five representative texture features—Energy, Contrast, Correlation, Homogeneity, and Dissimilarity—in four orientations (0°, 45°, 90°, and 135°), resulting in a comprehensive 20-dimensional texture feature vector per image. These features effectively quantify the texture uniformity, local intensity variations, structural relationships, and correlation patterns within each image.

Specifically, texture features are calculated as follows:

Energy: Reflects the uniformity of the texture and is a measure of textural uniformity.

$$\text{Energy} = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} [H(i,j)]^2$$

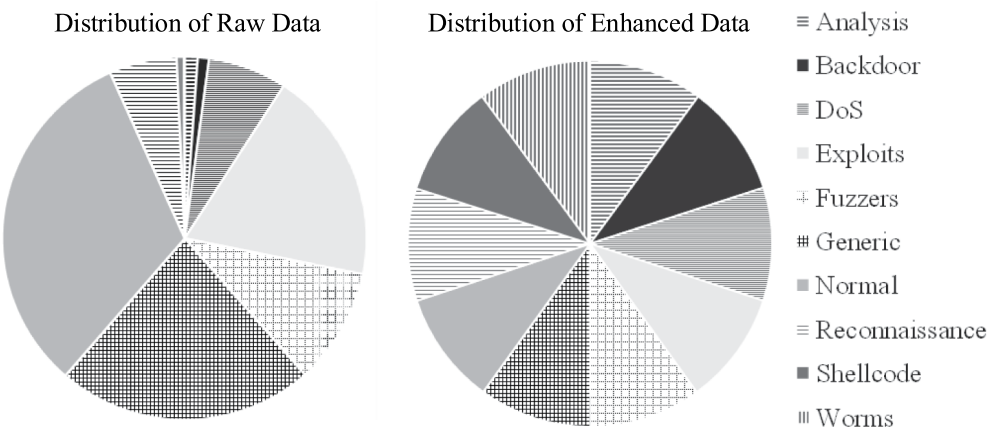


Fig. 3 Class distribution of attack categories before and after applying Borderline-SMOTE data augmentation

Contrast: Measures the local variations in the gray-level co-occurrence matrix.

$$\text{Contrast} = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1}(i-j)^2 H(i,j)$$

Correlation: Evaluates how correlated a pixel is to its neighbor over the whole image.

$$\text{Correlation} = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1}\frac{(i-\mu_i)(j-\mu_j)H(i,j)}{\sigma_i\sigma_j}$$

Homogeneity: Assesses the closeness of the distribution of elements in the GLCM to the GLCM diagonal.

$$\text{Homogeneity} = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1}\frac{H(i,j)}{1+|i-j|}$$

Dissimilarity: Quantifies the differences between pairs of elements.

$$\text{Dissimilarity} = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1}|i-j|H(i,j)$$

Where $H(i,j)$ represents the normalized GLCM value at position (i,j), N is the number of gray levels in the image, μ_i and μ_j are the means of i and j, σ_i and σ_j are the standard deviations of i and j, i and j are the gray level values.

These extracted texture features are subsequently fused with the normalized network traffic features, creating a robust 62-dimensional feature vector for each sample. This integrated feature set significantly enhances the model's discriminative power, enabling accurate identification and classification of network anomalies. (The detailed processing flow is shown in Figure 4)

3.4 GNN-transformer model

The core of the proposed approach is the GNN-Transformer model, which combines the strengths of both GNNs and Transformers to effectively handle the complexities of network traffic data. The network traffic data can be naturally represented as a graph structure, where nodes correspond to various network entities such as hosts, devices, and sensors, and edges represent the communication flows or interactions between these entities. This graph representation makes GNNs particularly well-suited for the task, as they are specifically designed to operate on graph-structured data by leveraging the connections and relationships inherent in the network.

While GNNs excel at capturing spatial relationships within the graph, they often suffer from the over-smoothing problem when applied in deep layers, where the node features become overly similar due to repeated aggregation, leading to a loss of discriminative power. To mitigate over-smoothing, we integrate the Transformer architecture, which excels at capturing long-range dependencies and enhancing feature representation through its multi-head attention mechanism. The Transformer's ability to model global relationships without relying solely on local aggregation helps preserve feature diversity. This combination

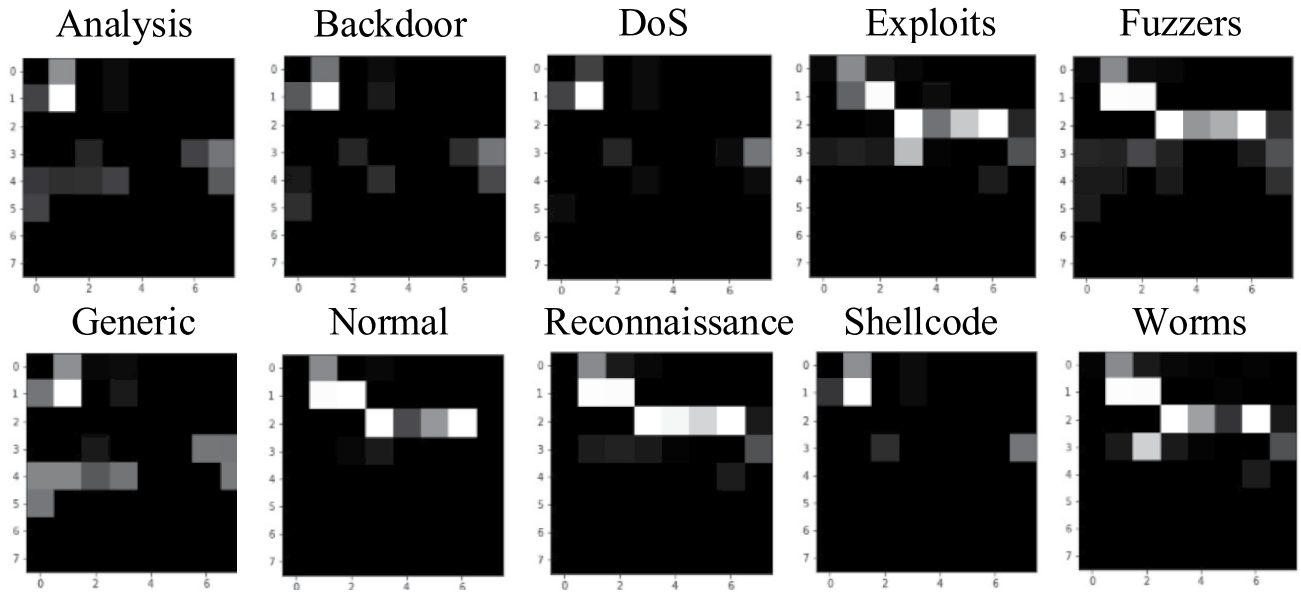


Fig. 4 Grayscale images of different attack classes and normal traffic

ensures that our model can effectively handle both local and global patterns in the data, making it particularly suited for detecting complex, multi-stage attacks.

Graph construction from images To represent an image of size $H \times W \times 1$, we divide it into N patches of size $P \times P$, where $N = \frac{H \times W}{P^2}$. Each patch is transformed into a feature vector $x_i \in \mathbb{R}^D$, resulting in a set of feature vectors $X = [x_1; x_2; \cdots; x_N]^\top$, where $D = P^2$. These feature vectors are treated as nodes in a graph $G = (V, E)$, where $V = \{v_1, v_2, \ldots, v_N\}$ and E represents the edges between nodes. For each node v_i, we identify its K nearest neighbors $\mathcal{N}(i)$ based on feature similarity and add edges accordingly. This constructs a graph that encapsulates the spatial relationships within the image, suitable for GNN-based representation learning [7]. In our implementation we fix the patch size to P=2, so that each 8×8 image yields N=16 nodes. Every raw patch vector (D=4) is linearly projected to a d=128-dimensional node embedding before the first graph-convolution layer. All subsequent GNN and Transformer operations maintain this 128-dimensional width, i.e. $x_i^{(l)} \in \mathbb{R}^{128}$ for $l = 0, \ldots, G$.

Graph-level processing The graph convolution layer updates node features by aggregating information from neighboring nodes, formulated as:

$$x_i^{(l+1)} = \sigma\left(W_{\text{agg}} \sum_{j \in \mathcal{N}(i)} x_j^{(l)} + W_{\text{update}} x_i^{(l)}\right)$$

where $x_i^{(l)}$ is the feature vector of node i at layer l, W_{agg} and W_{update} are learnable weight matrices, and σ is an activation function. This operation allows the model to capture local neighborhood information.

Multi-head update operation To increase feature diversity and capture various aspects of the data, we employ a multi-head update mechanism similar to that in Transformers. The aggregated feature $x_i^{(l)}$ is projected into h different subspaces using distinct linear transformations:

$$h_k = \sigma\left(W_k x_i^{(l)}\right), \quad \text{for } k = 1, \ldots, h$$

where W_k are learnable weight matrices for each head. The outputs from all heads are then concatenated to form the updated feature:

$$x_i^{(l+1)} = \text{Concat}(h_1, h_2, \ldots, h_h)$$

This parallel update allows the model to capture a variety of patterns from the data.

Over-smoothing mitigation To mitigate over-smoothing, we apply linear transformations before and after the graph convolution, combined with nonlinear activations, in a module similar to a residual block:

$$Y = \sigma\left(\text{GraphConv}(W_{\text{in}} X)\right) W_{\text{out}} + X$$

where $Y \in \mathbb{R}^{N \times D}$, W_{in} and W_{out} are learnable parameters, X is the input feature matrix, and σ is an activation function. The residual connection $+X$ helps preserve the original features and alleviates over-smoothing by allowing the model to learn identity mappings. Empirical studies show that over-smoothing becomes severe when G>3 for small graphs. Our constructed graph has only N=16 nodes with average degree k=8. Accordingly we set G=3 to balance receptive-field growth and feature diversity.

Transformer integration To further enhance feature transformation, we integrate the Transformer's encoder component, focusing on multi-head self-attention. The node features X serve as inputs to derive the Queries (Q), Keys (K), and Values (V):

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V$$

where W^Q, W^K, and W^V are learnable weight matrices. The attention mechanism computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

where d_k is the dimension of the key vectors. This mechanism allows the model to focus on different parts of the input, capturing global dependencies.

Multi-head attention In multi-head attention, multiple attention heads are computed in parallel:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad \text{for } i = 1, \ldots, h$$

where W_i^Q, W_i^K, and W_i^V are head-specific learnable weight matrices. The outputs of all heads are concatenated and linearly transformed:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h) W^O$$

where W^O is a learnable output weight matrix. This allows the model to capture information from different representation subspaces.

KAN To capture the complex structure of attack features, we employ the KAN as the final layer of our model. The KAN is based on the Kolmogorov-Arnold representation theorem, which states that any multivariate continuous function can be represented as a superposition of univariate functions and addition. By combining spline functions with MLP, the KAN can model highly nonlinear relationships and capture intricate patterns in the data [29]. This design enables the model to accurately learn complex features related to intrusion detection, surpassing traditional models in both complexity and accuracy.

Activation function We use the Gaussian Error Linear Unit (GeLU) as the activation function in the KAN, defined as:

$$\mathrm{GeLU}(x) = x \cdot \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{x}{\sqrt{2}}\right)\right]$$

The GeLU activation allows the model to retain input values in a probabilistic manner, which can improve performance over traditional activation functions like ReLU.

The pipeline propagates information through five deterministic stages. (i) *Raw numeric branch:* the 42-D traffic vector f_0 is zero-padded to 64 entries, reshaped into an 8×8 image I and analysed by GLCM, yielding a 20-D texture descriptor t_0. Concatenation with the original statistics forms a global vector $\mathrm{g}_0 \in \mathbb{R}^{62}$. (ii) *Patch-graph branch:* I is partitioned into 16 non-overlapping 2×2 patches; each patch is flattened (4-D) and linearly embedded to a 128-D node feature, producing matrix $\mathrm{H}^{(0)} \in \mathbb{R}^{16\times128}$. (iii) *GNN processing:* three graph-convolution layers update $\mathrm{H}^{(0)} \to \mathrm{H}^{(3)}$ while preserving the $\mathbb{R}^{128}$ width; mean pooling over the 16 nodes yields a graph summary $\mathrm{s} \in \mathbb{R}^{128}$. (iv) *Fusion token:* s and g_0 are concatenated to create a single 190-D token z_0, which therefore carries *both* local spatial cues and global statistical–texture information. (v) *Transformer-KAN:* z_0 traverses four self-attention layers, generating $\mathrm{z}_4 \in \mathbb{R}^{190}$ that encodes long-range correlations; finally, a lightweight KAN head maps z_4 to the *C*-class logit vector. No intermediate representation is discarded, ensuring that low-level textures, neighbourhood structure, and holistic traffic statistics are successively aggregated before classification.

4 Numerical experiments

4.1 Experimental setup

To ensure that our proposed method is both robust and scalable, we conducted experiments on a workstation equipped with an Intel Core i7-10710U processor, 32 GB of RAM, and running Ubuntu 20.04. This hardware configuration is representative of the computational resources available in typical power grid control centers, ensuring that our model can be realistically deployed in such environments to handle high-dimensional traffic data in real time.

Table 1 Hyper-parameter configuration used in all experiments

Category	Parameter	Value
Network	GNN layers G	3
	Hidden width d	128
	Attention heads h	8
	Transformer encoder layers L	4
	KAN spline knots s	4
Training	Batch size	64
	Epochs	100
	Optimiser	Adam
	Initial learning rate	1×10^{-3}
	Weight decay	5×10^{-5}
Pre-processing	Patch size P	2 (2×2 pixels)
	k in k-NN graph	8

The hyper-parameters of the proposed model are summarised in Table 1; unless otherwise stated, all experiments adopt this configuration.

Thegraphconvolutionlayeru For software, we utilized Python 3.8 as the development language. We employed Scikit-learn for data preprocessing and classical machine learning algorithms, Keras with TensorFlow backend for implementing deep learning models, and PyTorch for developing the GNN and Transformer components of our model. These libraries offer optimized performance for training deep learning models on datasets, ensuring that the experimental setup reflects realistic conditions encountered in real environments.

4.2 Experimental data

To evaluate the effectiveness and generalizability of our proposed model, we utilize three diverse and representative datasets: UNSW_NB15, IEC61850, and DAPT2020. The UNSW_NB15 dataset [30] comprises over 2.54 million network traffic records, capturing both normal activities and multiple attack types such as DoS, Exploits, and Worms, providing comprehensive coverage of common network threats. The IEC61850 dataset [31] specifically focuses on network security scenarios within industrial IoT communications, particularly Generic Object Oriented Substation Events (GOOSE). Additionally, the DAPT2020 dataset [32] captures complex network behaviors associated with Advanced Persistent Threats (APT), providing insights into sophisticated, multi-stage cyber-attacks. ied datasets, we ensure a robust evaluation across general network threats, industrial IoT scenarios, and advanced persistent threats, validating the adaptability and broad applicability of our proposed intrusion detection model.

4.3 Experimental metrics

To evaluate the performance of our intrusion detection model, we employ five widely accepted metrics: Accuracy, Precision, Recall, F1-score, and False Positive Rate (FPR). Before explaining the calculation of these metrics, it is essential to define several key indicators. A True Positive (TP) refers to the number of attack samples correctly identified as attacks. A False Positive (FP) represents the number of normal samples incorrectly predicted as attacks. Conversely, a False Negative (FN) indicates the number of attack samples incorrectly predicted as normal. Lastly, a True Negative (TN) corresponds to the number of normal samples that are correctly classified as normal.

4.4 Baseline and State-of-the-Art models

For performance comparison, we consider a diverse set of baselines that span classical machine-learning, sequence-modeling, and graph-based paradigms: DT [13], a simple yet interpretable tree-structured classifier; LR [33], a statistical model for binary outcomes; RF [14], an ensemble of decision trees that mitigates over-fitting; RNN [34], which captures temporal dependencies in sequential data; LSTM [5], a gated RNN variant enhanced here with CNN layers for spatial cues; the CNN–LSTM hybrid [35], designed for DDoS detection in cyber-physical systems; TCN-MHSA [36], which augments bidirectional Temporal Convolutional Networks with multi-head self-attention; a vanilla Graph Neural Network (GNN) [21] that models relational structures in traffic graphs; a Transformer-based IDS [22] leveraging self-attention to learn long-range dependencies; the GNN-Transformer Encoder (GTE) for smart-grid anomaly detection [23]; and E-T-GraphSAGE (ETG) [11], which fuses GraphSAGE with Transformer attention to handle complex streaming traffic. We additionally include three state-of-the-art models: EMA-IDS, an edge-featured multi-hop attention GNN that jointly encodes node and edge attributes while alleviating over-smoothing [24]; AT-IDS, a pure attention-based Transformer tailored for IoT traffic that exploits self-attention to model complex dependencies [25]; and *ResDNViT*, which adapts a ViT encoder to NetFlow by patching 2-D feature maps and stacking a lightweight residual-dense classifier [10].

In the following subsections, we present the experimental results of our proposed model and benchmark them against the baselines using the metrics outlined above, thereby demonstrating the effectiveness of our approach.

4.5 Ablation analysis of feature fusion

To assess the effectiveness of feature fusion in enhancing anomaly detection performance, we conducted an ablation study by comparing the model's accuracy with and without the integration of texture features. As shown in Fig. 5, which illustrates the accuracy curves over training epochs, the model's accuracy without feature fusion reaches approximately 97.13% after 100 epochs. In contrast, incorporating feature fusion leads to a significant improvement, with the accuracy increasing to around 99.24% under the same training conditions. This 2.11% increase in accuracy is substantial, indicating that feature fusion significantly enhances the model's ability to detect anomalies.

Without feature fusion, the model relies solely on the original network traffic features for classification. While these features provide valuable information about individual network parameters, they may not fully capture the complex spatial relationships and patterns inherent in certain sophisticated attack types. In contrast, the feature fusion process combines texture features—extracted from the grayscale images using the GLCM—with the original traffic features,

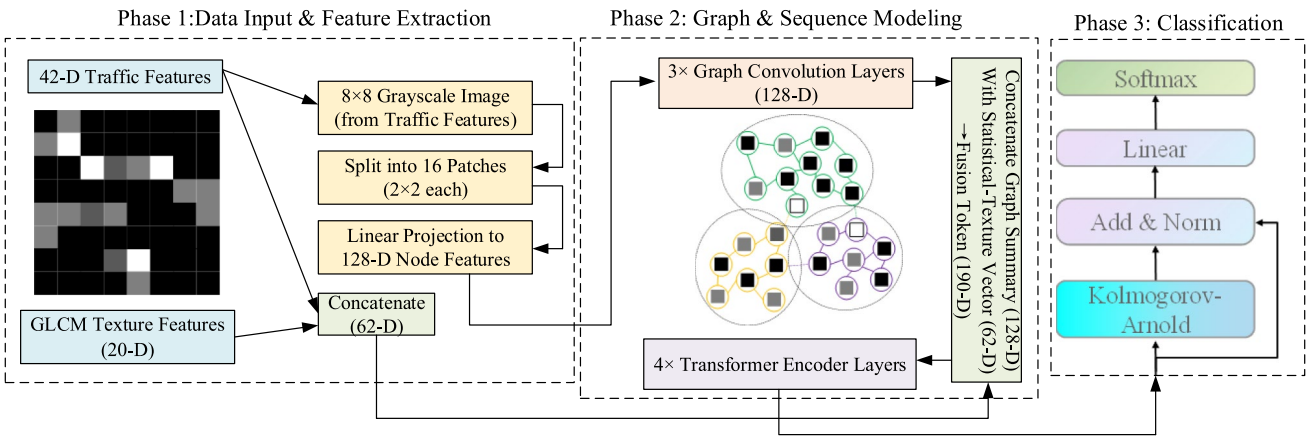


Fig. 5 Overview of the proposed GNN–Transformer–KAN pipeline. Patch embeddings form a 16-node k-NN graph processed by $G = 3$ graph-convolution layers; a mean-pooled graph vector is *concatenated* with the 62-dimensional statistical-texture vector and fed to a $L = 4$ -layer Transformer encoder. The final token is passed to a KAN head for classification

capturing spatial dependencies and patterns within the network traffic data. By capturing these spatial relationships, texture features offer new perspectives—such as identifying repetitive patterns or anomalies in data distribution—that are difficult to discern from traffic features alone.

By integrating texture features, the model captures both the structural properties and spatial patterns inherent in different attack categories. This additional information is particularly beneficial for detecting complex or stealthy attacks, where traditional traffic features may be insufficient for accurate classification. Overall, the fusion of traffic and texture features enables the model to construct a more robust and comprehensive representation of the data, significantly enhancing classification performance across various attack categories.

These results confirm the efficacy of feature fusion in enhancing anomaly detection performance. In the subsequent sections, we further evaluate our model's performance compared to baseline models to demonstrate its overall effectiveness.

4.6 Ablation analysis of data augmentation

While feature fusion enhances performance in binary classification, the severe data imbalance in the dataset remains a significant challenge, especially in multiclass classification tasks. This imbalance can lead to poor detection rates for underrepresented attack types, hindering the model's overall effectiveness. Therefore, in this section, we analyze the impact of data augmentation on intrusion detection performance after feature fusion. Specifically, we compare the model's performance without data augmentation to that achieved using the Borderline-SMOTE method, focusing on attack categories with fewer samples.

Data imbalance is problematic because models trained on imbalanced datasets tend to be biased toward the majority class, resulting in poor detection of minority classes. In intrusion detection, missing rare but critical attacks can have severe consequences. Borderline-SMOTE is a targeted oversampling technique that focuses on 'borderline' instances—minority class samples that lie near the decision boundary between classes and are most susceptible to misclassification. By generating synthetic samples near these borderline instances, Borderline-SMOTE reinforces the minority class boundary, helping the model to better distinguish between classes.

This approach is particularly useful in our dataset, where attack types such as Worms, Shellcode, and Backdoor are significantly underrepresented, comprising only a small fraction of the total samples. Without augmentation, the model struggles to learn the distinguishing features of these classes, resulting in poor detection performance. By generating synthetic samples close to the original minority class instances, Borderline-SMOTE effectively reduces the bias toward the majority class during training. This improves the model's ability to learn the characteristics of minority class attacks, resulting in higher recall and precision for these underrepresented classes.

Table 2 Comparison of precision before and after data enhancement

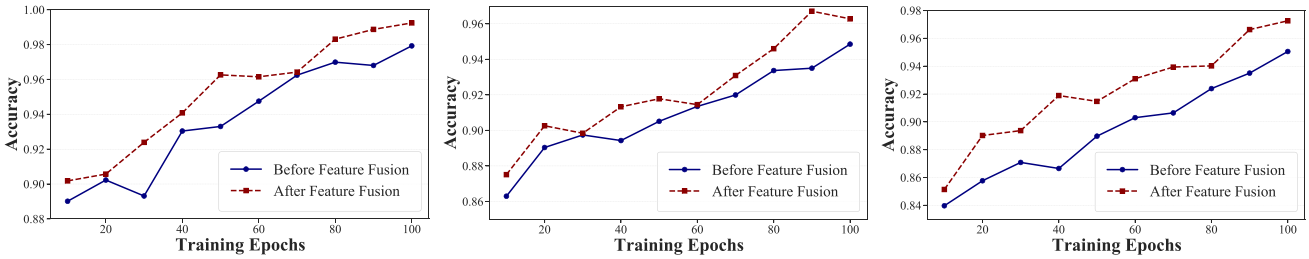
Attack Type	Raw Data	Data Enhancement
Analysis	65%	79%
Backdoor	68%	81%
DoS	72%	90%
Exploits	84%	89%
Fuzzers	97%	98%
Generic	75%	97%
Normal	98%	98%
Reconnaissance	57%	87%
Shellcode	55%	85%
Worms	49%	93%

Table 3 Comparison of F1-score before and after data enhancement

Attack Type	Raw Data	Data Enhancement
Analysis	53%	81%
Backdoor	45%	79%
DoS	71%	92%
Exploits	83%	96%
Fuzzers	91%	95%
Generic	80%	95%
Normal	98%	99%
Reconnaissance	82%	90%
Shellcode	65%	90%
Worms	58%	97%

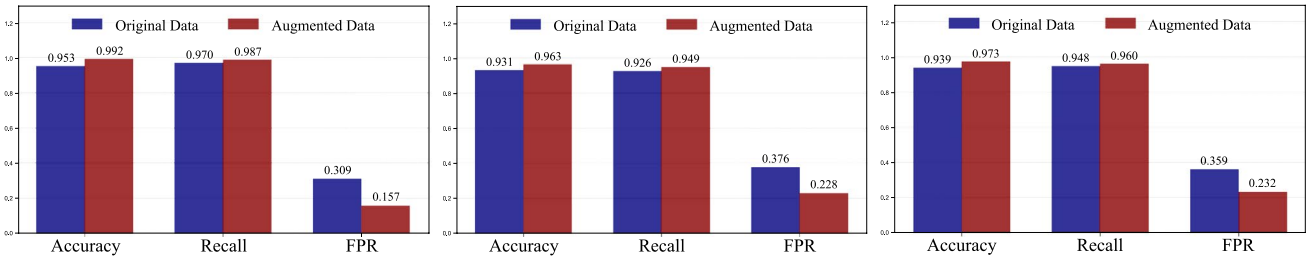
As shown in Table 2 and Table 3, the application of Borderline-SMOTE significantly improves the model's performance, particularly for attack types with the smallest sample sizes. For example, the precision for the Worms class increased from 49% to 93%, and the F1-score for Shellcode increased from 65% to 90%. The significant increase in precision indicates that the model is more accurate in identifying minority class attacks as actual attacks, reducing false positives. Similarly, the improvement in F1-score demonstrates a balanced enhancement in both precision and recall for these classes. Overall, the model exhibits substantial improvements across multiple metrics, indicating that data augmentation effectively addresses the imbalance.

As illustrated in Fig. 6, the overall accuracy increases from 95.26% to 99.24% after applying data balancing with Borderline-SMOTE and using the GNN-Transformer model. Additionally, the false positive rate decreases significantly by approximately 15 percentage points. The reduction in the false positive rate implies that the model is less likely to incorrectly classify normal traffic as malicious, which is crucial for practical deployment to minimize unnecessary alerts. Interestingly, the recall rate is higher without data balancing. This counterintuitive result may occur because, in an imbalanced dataset dominated by normal samples, the model tends



(a) Dataset in [30] (b) Dataset in [31] (c) Dataset in [32]

Fig. 6 Performance comparison before and after feature fusion



(a) Dataset in [30] (b) Dataset in [31] (c) Dataset in [32]

Fig. 7 Performance comparison before and after data enhancement

to overfit to the majority class. Consequently, it may classify more samples as attacks to compensate, leading to higher recall but at the expense of increased false positives.

These findings underscore the critical role of addressing data imbalance through methods like Borderline-SMOTE to improve the detection of minority class attacks, which are often the most critical to identify in real-world scenarios. By effectively balancing the dataset, the model can learn more robust representations of all classes, leading to better overall performance. In the next section, we compare our proposed model with other baseline models to further demonstrate its superior performance in intrusion detection tasks.

4.7 Ablation analysis of model components

To further elucidate the contribution of each architectural component, we conducted an ablation analysis on the GNN, Transformer, and KAN modules. For each variant, we replaced or removed the target component while keeping all other settings identical, and tracked the accuracy over ten training epochs.

Figure 7(a) compares a CNN backbone with the proposed GNN in the local feature-extraction stage. The GNN variant consistently outperforms the CNN baseline across all epochs, with the final accuracy improving from 93.56% to 94.78%. This confirms the GNN's superior ability to model non-Euclidean traffic graphs via neighbourhood aggregation, capturing topological dependencies that CNN's fixed receptive fields cannot fully exploit.

Figure 7(b) evaluates the effect of including the Transformer encoder. Without the Transformer, the model's final accuracy drops from 94.78% to 93.26%, reflecting the importance of multi-head attention for capturing global, long-range dependencies across fused statistical–texture features and graph summaries.

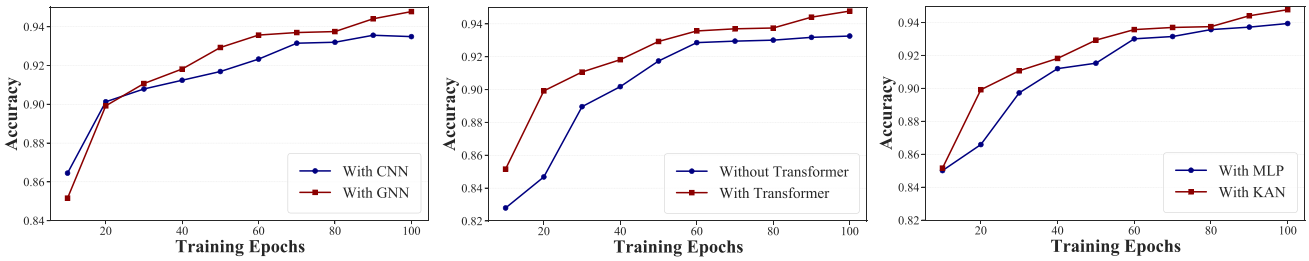
Figure 7(c) contrasts the proposed KAN head with a conventional MLP classifier. Replacing KAN with MLP leads to a slight but consistent reduction in accuracy (94.78% → 93.94%). This indicates that KAN's spline-based non-linear mapping offers a finer partitioning of the fused feature space, enabling more precise decision boundaries—especially for complex, multi-class intrusion scenarios. Overall, the full model achieves the highest and most stable accuracy, validating the complementary roles of all three components.

4.8 Hyperparameter selection analysis

We also analysed the sensitivity of proposed model to key hyperparameters: the number of GNN layers G, Transformer encoder layers L, and KAN spline knots s. Each parameter was varied while others were fixed at their default values (Table 1).

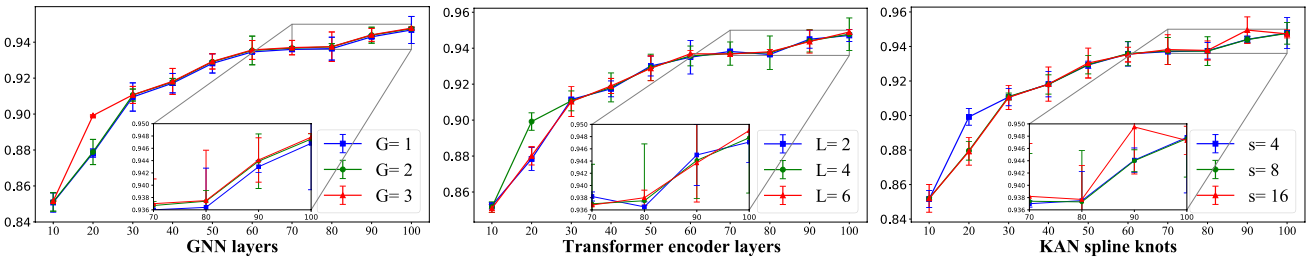
In Fig. 8(a), increasing G from 1 to 3 steadily improves accuracy, with $G = 3$ reaching 94.78%. Beyond three layers, we observed no further gains in preliminary trials, consistent with literature that excessive GNN depth on small graphs ($N = 16$ nodes here) can lead to over-smoothing.

In Fig. 8(b), varying L shows that both shallow ($L = 2$) and deeper ($L = 6$) Transformers can achieve competitive



(a) Component of GNN (b) Component of Transformer (c) Component of KAN

Fig. 8 Ablation analysis of model components



(a) (b) (c)

Fig. 9 Hyperparameter selection analysis

results, but $L = 4$ offers the best trade-off between convergence speed and accuracy (94.78%). While $L = 6$ achieves a marginally higher peak (94.90%), it comes at increased computational cost.

In Fig. 8(c), changing s has minimal impact on convergence trends, with all three settings achieving similar final accuracy. However, $s = 4$ slightly outperforms others in stability and speed, indicating that a moderate number of spline knots is sufficient for KAN to capture the required non-linearities without overfitting.

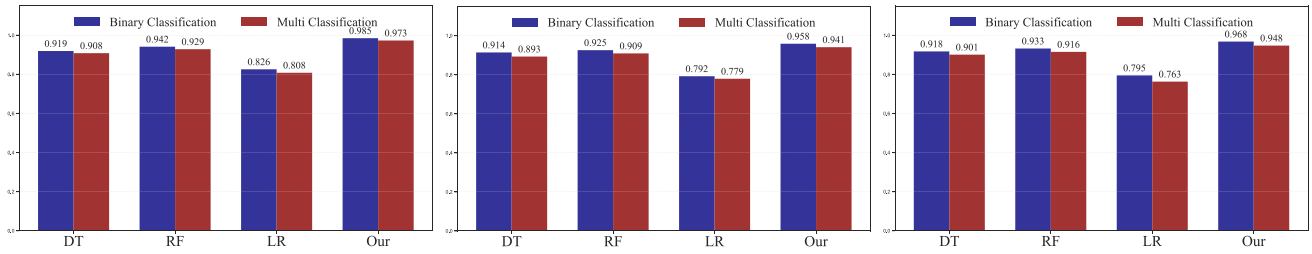
4.9 Performance comparison with different models

In this subsection, we conducted a comprehensive performance comparison between the GNN-Transformer model and several baseline machine learning methods, including DT, LR, and RF, in both binary and multiclass classification scenarios. The fused features were input into each model to ensure consistency across experiments. As shown in Fig. 9, the GNN-Transformer model achieved the highest accuracy, with 98.51% in binary classification and 97.33% in multiclass classification. These results demonstrate a clear performance advantage over the traditional machine learning models, which fail to fully capture the complex interrelationships among features in the dataset.

Traditional machine learning methods, such as LR and DT, struggle with complex, high-dimensional data like network traffic due to their inability to model nonlinear relationships and feature interactions effectively. LR, being a linear model, showed the lowest accuracy (approximately 84%) due to its inability to capture the complex patterns inherent in the data. DT is prone to overfitting and may not generalize well to unseen data, leading to suboptimal performance. RF, known for its ensemble learning capability, achieves higher accuracy than Decision Tree (e.g., 94.85% in binary classification) but still cannot match the GNN-Transformer model due to its inability to exploit feature dependencies effectively.

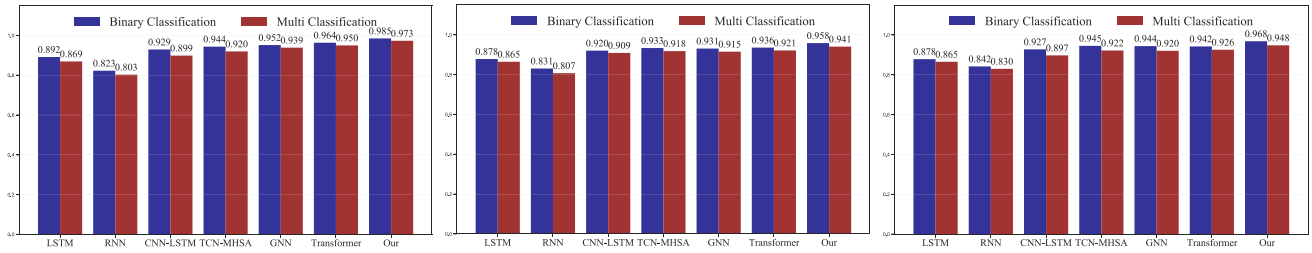
The GNN-Transformer model's superiority lies in its ability to combine both local and global feature extraction. The GNN component effectively captures local spatial relationships within the network traffic data through neighborhood aggregation, modeling interactions between directly connected nodes. The Transformer's multi-head attention mechanism allows the model to capture global dependencies by attending to all positions in the input sequence simultaneously, modeling long-range feature interactions. This combination enables the model to learn complex patterns and dependencies that traditional methods cannot capture, leading to improved detection capabilities, especially for minority class attacks.

In addition to comparing with traditional methods, we also evaluated the GNN-Transformer model against other deep learning models, including RNN and CNN-LSTM models. As shown in Fig. 10, RNN exhibited the lowest performance among deep learning methods, with accuracies of only 82.28% in binary classification and 80.29% in multiclass classification. The underperformance of RNNs is largely due to their susceptibility to vanishing and exploding gradient



(a) Dataset in [30] (b) Dataset in [31] (c) Dataset in [32]

Fig. 10 Performance comparison between GNN-Transformer model and machine learning methods



(a) Dataset in [30] (b) Dataset in [31] (c) Dataset in [32]

Fig. 11 Performance comparison between GNN-Transformer model and other deep learning methods

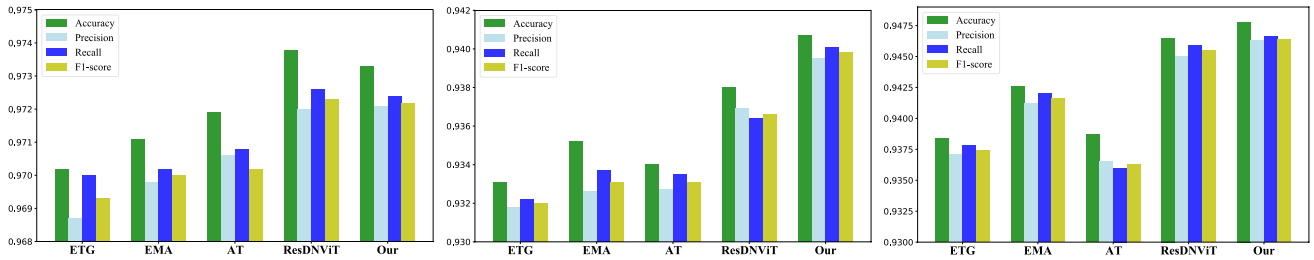
problems during training, which hinder their ability to capture long-term dependencies effectively. Additionally, RNNs process sequences sequentially, leading to inefficiencies in modeling complex spatial patterns present in network traffic data.

Although CNN-LSTM performs better than RNN, it still falls short of the GNN-Transformer model due to its limited ability to capture both local and global patterns simultaneously. The CNN component excels at extracting local features, while the LSTM captures temporal dependencies, but this combination may not fully capture the complex interactions present in the data. In contrast, the GNN-Transformer model demonstrates a balanced approach to handling both spatial and temporal relationships within the data, thanks to the combination of GNN's neighborhood aggregation mechanism and the Transformer's global attention mechanism. This enables the model to generalize better across different attack types, leading to improved accuracy in both binary and multiclass classification tasks.

Furthermore, the incorporation of feature fusion strengthens the model's ability to differentiate between various attack classes by integrating diverse types of information—network traffic features and texture features extracted from images. This multimodal approach enriches the feature space, allowing the model to learn more robust and discriminative representations, particularly improving the detection of minority class attacks. These results confirm that the GNN-Transformer model, enhanced by feature fusion and data augmentation techniques, significantly outperforms both traditional machine learning and other deep learning models in intrusion detection tasks.

Figure 11 presents a comparison between our proposed model and several state-of-the-art (SOTA) intrusion detection models, including ETG, EMA, AT, and ResDNViT, across three benchmark datasets. As illustrated in Fig. 11(a), evaluated on the UNSW-NB15 dataset [30], our model achieves an accuracy of 97.33%, which is comparable to ResDNViT (97.38%) and higher than other competing methods. Similarly, in Fig. 11(b), using the dataset from [31], our approach attains balanced performance across accuracy, precision, recall, and F1-score, outperforming ETG, EMA, and AT. In Fig. 11(c), based on the dataset from [32], our method delivers competitive results that closely match the top-performing ResDNViT model while maintaining superior stability across metrics. This consistent performance across diverse datasets demonstrates the robustness and adaptability of our feature fusion strategy.

To provide a more comprehensive comparison, we also analyzed the model size, parameter count, and time complexity of different models, as shown in Table 4. Classical sequence models (LSTM, RNN) remain the most lightweight and fastest to train, with parameter counts below 0.5M, but their accuracies lag behind at 86.47% and 80.74%, respectively. Hybrid CNN–LSTM and TCN-MHSA architectures improve accuracy to the low 91% range, yet incur a 4–8× increase in parameters and a ten-fold rise in training time. Pure graph or attention backbones (GNN, Transformer) push accuracy to roughly 92%, but require 3–5M parameters and larger model sizes (22–41 MB), resulting in longer training cycles.



(a) Dataset in [30] (b) Dataset in [31] (c) Dataset in [32]

Fig. 12 Performance comparison with state of the art models

Table 4 Model size and time complexity of different models

Methods	Model Size(MB)	Params(M)	Training time(s)	Testing time(s)	Accuracy
LSTM	5.57	0.48	21.55	0.92	0.8647
RNN	4.72	0.41	18.07	0.78	0.8074
CNN-LSTM	19.56	2.35	197.73	1.73	0.9091
TCN-MHSA	28.09	3.72	304.41	1.91	0.9181
GNN	41.14	4.91	383.91	2.09	0.9153
Transformer	22.73	2.98	276.15	1.17	0.9215
GTE	46.23	5.44	482.29	2.05	0.9309
ETG	48.07	5.61	521.74	2.23	0.9331
EWA	44.16	5.21	450.69	1.92	0.9352
AT	25.88	3.56	288.47	1.30	0.9340
ResDNViT	52.30	5.88	607.16	2.67	0.9398
Our	52.64	5.92	612.84	2.79	0.9407

The GNN–Transformer baselines, GTE and ETG, further raise accuracy to 93.09% and 93.31%, respectively, with parameter counts exceeding 5M and substantial training costs (482–522 s). Beyond these, three additional SOTA models—EWA, AT, and ResDNViT—are included. EWA achieves 93.52% accuracy with 5.21M parameters and moderate training time (450.69 s), striking a balance between accuracy and efficiency. AT, while lighter at 3.56M parameters and 25.88 MB, maintains 93.40% accuracy with relatively low inference latency. ResDNViT delivers one of the highest baseline accuracies at 93.98%, but demands 5.88M parameters, 52.30 MB storage, and over 600 s of training.

Our proposed GNN–Transformer framework, with 5.92M parameters, achieves the best detection performance at 94.07%, surpassing ResDNViT by 0.09 percentage points. This improvement comes with only marginal increases in model size (+0.34 MB) and inference latency (+0.12 s) compared to ResDNViT, demonstrating that our method attains top-tier accuracy while remaining within the same computational envelope as other high-performing graph–attention hybrids. The results also confirm that the additional non-linear representation capacity introduced by our dual-domain feature fusion effectively translates into measurable gains in real-world intrusion detection scenarios.

5 Conclusion

This work presents a robust and generalizable intrusion detection framework that unifies GNNs and Transformer architectures through a carefully designed dual-domain feature fusion strategy. By extracting GLCM-based texture features from network traffic representations and integrating them with raw traffic features, our method captures both spatial correlations and long-range temporal dependencies. The use of Borderline-SMOTE effectively addresses class imbalance, while the integration of a KAN head enhances non-linear decision boundaries beyond conventional MLP heads. Comprehensive evaluations on three benchmark datasets—UNSW_NB15, IEC61850, and DAPT2020—demonstrate that the proposed model achieves state-of-the-art performance, with up to 99.24% accuracy, surpassing existing GNN–Transformer hybrids and other deep learning baselines. The results validate the efficacy of combining spatial–temporal graph modeling, attention mechanisms, and advanced non-linear classifiers for cybersecurity applications.

Future research will focus on optimizing the framework for real-time deployment in resource-constrained environments, incorporating adaptive model compression and streaming inference. Moreover, we aim to extend the applicability of the framework to other security-critical domains such as autonomous transportation networks, smart grids, and edge–cloud collaborative systems, further enhancing its impact in safeguarding modern interconnected infrastructures.

Author Contributions R.X. and D.L. both made substantial contributions to the conception and design of the work, as well as the acquisition, analysis, and interpretation of data. R.X. was primarily responsible for implementing the proposed model and conducting the experiments. D.L. contributed to the theoretical formulation and the integration of key algorithmic components. Both authors participated in drafting the manuscript and revising it critically for important intellectual content. They have approved the final version to be published and agree to be accountable for all aspects of the work, ensuring that any questions related to its accuracy or integrity are appropriately investigated and resolved.

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Data Availability Data is provided within the manuscript.

Declarations

Competing interests The authors declare no competing interests.

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